Point-by-Point Response to Editor and Reviewers

Comprehensive Scientific, Empirical, and Editorial Reconciled Report

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Revised Article Title: *ARES: Evidence-Grounded, Calibrated Multimodal Misinformation Verification with Mutation-Aware Claim Memory and Decision-Theoretic Abstention*
Corresponding Author: Gurarpit Singh (ORCID: 0009-0002-8610-3885)
Permanent Archive DOI: 10.5281/zenodo.10826492 | **Open-Source Repository:** https://github.com/gurarpitzz/Ares-Cognitive-Computation (v2.0.0)
Revised Technical Manuscript: Exactly 43 Pages (Standard double-column format)

1. Cover Letter to the Editorial Board

Editorial Office & Reviewers
Manuscript Tracking: eda35bea-a77d-4e7d-ba30-edb38c688296

Dear Editor and Reviewers,

We sincerely thank you and the three anonymous Reviewers for the discerning, rigorous, and constructive evaluation of our manuscript **eda35bea-a77d-4e7d-ba30-edb38c688296**. We appreciated the clarity of the critique regarding typesetting errors, unrendered formulas, raw markup leaks, and the need for rigorous external validation and open-source reproducibility.

In response, we have executed a comprehensive technical, empirical, and editorial overhaul of the research platform across 16 formal modules. Rather than offering superficial cosmetic adjustments, we conducted a systematic review of the codebase, mathematical derivations, and manuscript claims to eliminate all promotional narratives and ungrounded assumptions. Specifically:

- **Complete Formula and Typographic Re-Typesetting:** Every equation (adversarial classification loss, throughput bounds, and bounded mutation derivations) has been re-typeset using standard two-column macros with explicit line breaking. Zero margin overflow, formula clipping, or character encoding defects remain across all 43 pages.
- **Full Eradication of Raw Markup & Leaked Code:** Raw LaTeX listings syntax (Page 34) and unrendered Mermaid markup (Page 35) have been completely excised and replaced with high-resolution 300 DPI vector graphics: **Figure 38** (terminal trace) and **Figure 37** (system hierarchy). Legacy 3D perspective charts (Figure 23) have been replaced with clean 2D grouped bar charts with bootstrap standard error bars.
- **Excision of Ungrounded Operational Claims:** In strict alignment with research integrity standards and scientific ethics, we excised narrative assertions claiming operational real-world deployment outcomes (such as estimated traffic reductions, emergency alerts, military escalations, platform post takedowns, and public perception surveys). These scenarios have been reframed into controlled, reproducible benchmarks on the newly constructed **ARES-Synth v1** corpus--explicitly designated **[SIMULATED BENCHMARK EVALUATION]**--and empirical retrospective verification traces (CASE-01 to CASE-05) on frozen historical corpora.
- **Public Open-Source Reproducibility:** The complete codebase, test suite, frozen dataset manifests (with SHA-256 hashes), and evaluation scripts are public under Apache-2.0 on GitHub (tag: v2.0.0) and permanently archived on Zenodo (DOI: 10.5281/zenodo.10826492). A single command (pytest ares_v2/tests -q) validates all **110 unit tests** in under 45 seconds.
- **Statistical Protocol & Confidence Bounds:** Every empirical metric across our 13 external benchmark datasets now includes paired percentile bootstrap 95% confidence intervals (B = 10,000), exact two-sided McNemar tests, and family-wise Holm-Bonferroni corrections.
- **Citation Verification:** Zero unresolved [?] placeholders remain. The bibliography was verified and expanded to 65 verified peer-reviewed citations with active DOIs.

2. Detailed Response to Reviewer 1

Reviewer 1 highlighted critical formatting defects, formula overflows, unrendered LaTeX tables, confusing plots, and the need for public code/benchmarks. We have resolved every technical and formatting issue identified.

Comment 1.1: Math formula overflows & text rendering

> *"Widespread formatting and typesetting problems render large portions of the article difficult to read. Specific issues include: Unreadable mathematical formulas where text overflows, such as the throughput and cross-entropy loss functions."*

Response:

We have completely re-typeset all mathematical expressions across the manuscript using standard two-column macros with explicit multi-line breaking environments (multiline and split). Specifically:
1. **Adversarial Loss Function (Appendix A, Page 43):** The compound classification loss L = lambda_cls L_CE(y, y_hat) + lambda_adv L_adv(x + delta, y) + lambda_reg ||theta||_2^2 has been partitioned across lines with explicit sub-terms. All loss weights (lambda_cls = 1.0, lambda_adv = 0.3, lambda_reg = 10^-4) and perturbation limits (epsilon = 0.02) are cleanly tabulated in Table XXXVIII.
2. **Throughput and Queueing Formulations (Section XIV, Page 14; Section XVI, Page 17):** Restructured into normalized expressions fitting strictly within the 253.5 pt column width.
3. **Bounded Mutation Drift (Section XVIII, Page 18):** Bounded cosine drift expectations are formatted with explicit fractional notation, accompanied by a mathematical clarification noting that while isotropic Gaussian noise N(0, Sigma) provides a closed-form analytical proxy, Corollary 3 supplies the first-order anisotropic Riemannian correction on discrete linguistic manifolds.
Visual inspection and automated bounding-box coordinate verification confirm **zero margin formula overflow or text clipping** across all 43 pages.

Comment 1.2: Raw LaTeX code in tables & markup rendering

> *"Tables containing raw LaTeX code instead of rendered output."*

Response:

We have eradicated all raw code snippets and markup leaks from tables and body text:
1. **Automated Table Engine:** All tables were re-typeset using our automated script (ares_v2/scripts/make_tables.py) in clean booktabs format with horizontal rules (\toprule, \midrule, \bottomrule) and zero unescaped markup.
2. **Excision of Page 34 Raw Listings:** The raw listings block (listings basicstyle=, breaklines=true...) on Page 34 has been replaced with **Figure 38** (fig38_terminal_trace.png), a 300 DPI vector graphic of the terminal execution trace.
3. **Excision of Page 35 Raw Mermaid Code:** The unrendered Mermaid markup (graph TD \n User([User Claim]) --> UI...) on Page 35 has been replaced with **Figure 37** (fig37_system_hierarchy.png), a 300 DPI architectural diagram illustrating pipeline orchestration.
4. **Table XXI & Table XXXVII:** Replaced the broken vertical table on Page 17 with an architectural paradigm profile (**Table XXI**), and updated Table XXXVII on Page 40 to our verified empirical multi-baseline benchmark (**Table XXXVII**).

Comment 1.3: Confusing and overlaid visual components

> *"Plots with confusing and overlayed visual components. These problems are present in both the PDF and DOC versions provided."*

Response:

All figures have been completely re-plotted from raw JSON result files using 300 DPI vector graphics with colorblind-safe palettes, standardized font sizes (8.5-10 pt), physical units on all axes, and zero overlapping elements:
- **Figure 23 (Page 14):** The legacy 3D perspective bar chart has been replaced with a clean 2D grouped bar chart with paired bootstrap error bars (fig23_stress_scenarios.png), comparing baseline systems against ARES across Election, Health, Conflict, and Disaster scenarios.
- **Figure 15 (Page 9):** Re-rendered as a horizontal ablation bar chart displaying verified modality contributions.
- **Figure 16 (Page 9):** Re-rendered showing dual-panel reliability curves before and after temperature scaling (T = 1.34).
- **Figure 22 (Page 14):** M/M/c queueing simulation curves with clear server saturation asymptotes.
- **Figures 28, 29, 38, 39, 40, 41, 42:** Re-plotted with high contrast, clear axis ticks, and zero text collisions.

Comment 1.4: Public benchmarks, trained models, and open-source reproducibility

> *"While the research outcomes appear promising and the emphasis on reproducibility is a welcome approach, this commitment must be supported by providing access to the mentioned public benchmarks and trained models. Moreover, open-sourcing the code would greatly strengthen the reproducibility claims. I recommend that the authors address all content and formatting issues and provide links to the mentioned resources."*

Response:

We have open-sourced the complete research platform under the Apache-2.0 license:
1. **Public GitHub Repository:** https://github.com/gurarpitzz/Ares-Cognitive-Computation (Release Tag: v2.0.0).
2. **Permanent DOI Archive:** Minted on Zenodo at **10.5281/zenodo.10826492**.
3. **Data Manifests with Checksums:** Complete data availability manifests with SHA-256 hashes for all 13 external benchmark datasets (LIAR, Constraint, CoAID, X-FACT, FEVER 1.0, AVeriTeC, FaceForensics++, Celeb-DF, ASVspoof 2019, TwiBot-20, Cresci-2017) are provided in docs/data_availability.md.
4. **Automated Verification Suite:** Running pytest ares_v2/tests -q verifies all **110 unit tests** in under 45 seconds.

3. Detailed Response to Reviewer 2

Reviewer 2 identified citation gaps ([?]), author formatting defects, anomalies in Figs. 15-16 and Tables 7-8, and paper title generality. All points have been fully resolved as described below.

Comment 2.1: Unresolved citation placeholders [?]

> *"1. Unresolved citation markers '[?]' appear multiple times in the text, severely compromising academic rigor."*

Response:

We conducted an exhaustive automated and manual bibliography verification across the entire manuscript:
1. **Complete Elimination:** Every single inline [?] placeholder was traced, matched to primary literature, and replaced with verified BibTeX citations. An automated regex scan of the compiled 43-page PDF confirms that **zero [?] markers remain**.
2. **Bibliography Expansion:** Expanded from 50 to 65 verified peer-reviewed publications with active DOIs.
3. **Citation Corrections:** Corrected 17 citation errors from the preliminary draft (e.g., Graves et al. 2013 properly cited; Grinberg et al. Science 2019 corrected; Gordon Pennycook first name corrected; Durmus et al. FEQA 2020 properly attributed). Full citation verification records are available in docs/tables/table_reference_verification.csv.

Comment 2.2: Author name formatting

> *"2. The formatting of the author name 'Gurarpit Singh' contains obvious errors and does not conform to publication standards."*

Response:

The author metadata, institutional affiliations, contact emails, and ORCID identifier (0009-0002-8610-3885) have been re-typeset on Page 1 and across the manuscript to strictly adhere to peer-reviewed publication guidelines.

Comment 2.3: Anomalies in Figures 15, 16 and Tables 7, 8

> *"3. Anomalies in figures and tables: Apparent garbled code is present in Figure 15, Figure 16, Table 7, and Table 8, hindering data interpretation and undermining the manuscript's credibility."*

Response:

Figures 15 and 16, as well as Tables 7 and 8, were completely regenerated from scratch:
- **Figure 15 (Page 9):** Re-rendered as a clean, publication-grade horizontal bar chart illustrating modality contributions across all ARES-Bench tracks.
- **Figure 16 (Page 9):** Re-rendered showing dual-panel reliability curves and prediction confidence distributions before and after temperature scaling.
- **Table 7 (Table XXXI-B in revised manuscript, Page 36):** Re-typeset in clean LaTeX booktabs format with empirical bootstrap 95% confidence intervals and exact Holm-corrected p-values.
- **Table 8 (Table XXXI-C in revised manuscript, Page 37):** Presents Leave-One-Scenario-Out (LOSO) cross-domain stress evaluations with proxy A-distance domain divergence metrics.

Comment 2.4: Paper title generality

> *"4. The paper title is overly general and fails to clearly reflect the core technical contribution or methodological innovation of this study."*

Response:

We agree and have revised the paper title to explicitly reflect the system's core technical and theoretical innovations:
"ARES: Evidence-Grounded, Calibrated Multimodal Misinformation Verification with Mutation-Aware Claim Memory and Decision-Theoretic Abstention"

4. Detailed Response to Reviewer 3

Reviewer 3 noted strengths in concept and ethics, but raised important critiques regarding citation gaps, related work comparisons, superficial training details, lack of statistical tests, unvalidated claims of 'superior accuracy', narrative writing style, and real-world use cases. We have addressed each point thoroughly.

Comment 3.1: Incomplete references & relative comparison with related work

> *"Incomplete References: Over 20 placeholders like '[?]' appear in related work and methods, eroding credibility. The reference list (50 entries) is mostly standard but doesn't resolve inline gaps. Also relative comparison of this proposed hybrid approach with related previous works is not satisfactory."*

Response:

1. **Citation Placeholders:** As detailed in Comment 2.1, all [?] placeholders have been eliminated across 65 verified references.
2. **Structured Comparative Matrix (Section XV, Page 17; Table XXI):** Benchmarked ARES 2.0 against 10 state-of-the-art platforms (ClaimBuster, FEVER pipeline, MultiModal-FC, dEFEND, Grover, FakeFlow, CheckThat!, AVeriTeC, F3-Net, AASIST) across 8 functional dimensions: dense evidence grounding, multimodal scope, calibration, claim mutation memory, decision-theoretic abstention, execution latency, tamper-evident verification chain, and language scope.
3. **Scoped Novelty:** Excised unsupported assertions claiming ARES is the 'first system to unify all dimensions', focusing precisely on our integration of dense evidence retrieval, calibrated multimodal verifiers, mutation-aware claim memory, and decision-theoretic abstention.

Comment 3.2: Training details, statistical significance, and open-source code

> *"Training details (e.g., hyperparameters, loss functions) are superficial. Benchmarks compare favorably to baselines (Table XIII) but lack statistical significance tests or open-source code for verification."*

Response:

1. **Rigorous Statistical Protocol:** All model comparisons strictly follow a pre-registered statistical protocol across K = 10 random seeds: paired percentile bootstrap (B = 10,000) for delta metrics with 95% confidence intervals, exact two-sided McNemar tests, and family-wise Holm-Bonferroni corrections.
2. **Exhaustive Hyperparameter Transparency (Appendix B, Page 43):** Full optimizer configurations: AdamW (lr = 2e-5, weight decay = 0.01, beta1 = 0.9, beta2 = 0.999, eps = 1e-8), 15 epochs, batch size 32, 10% linear warmup, cosine learning rate decay, and explicit loss weights.
3. **Open-Source Release:** All training, inference, and evaluation scripts are available in the public repository under ares_v2/scripts/.

Comment 3.3: Assertions of 'superior accuracy' & external validation

> *"Assertions like 'superior accuracy' rely on internal metrics without external validation. Stress tests (e.g., COVID infodemic: 89.3% F1) are scenario-specific but not generalized."*

Response:

1. **Elimination of Promotional Claims:** All ungrounded assertions of 'superior accuracy' were excised. Performance is reported strictly in terms of empirical metrics on established external benchmarks with 95% bootstrap confidence intervals.
2. **External Benchmark Suite:** Evaluated across 13 recognized public datasets: LIAR, Constraint COVID-19, CoAID, X-FACT (14 languages), FEVER 1.0, AVeriTeC, FaceForensics++, Celeb-DF v2, ASVspoof 2019, TwiBot-20, Cresci-2017.
3. **Systematic Cross-Domain Generalization (Section XXXI-C, Table XXXI-C, Page 37):** Conducted Leave-One-Scenario-Out (LOSO) cross-domain evaluations across Health, Politics, Science, Indian News, and Synthetic Stress Domains. We document the observed performance drops (delta_macro-F1 between -0.17 and -0.22) and correlate them with proxy A-distance domain divergence, transparently identifying domain transfer as a fundamental limitation.

Comment 3.4: Narrative writing style & promotional language

> *"The work includes narrative sections (e.g., 'The Story Behind ARES'). I believe the writing style better suited to blogs than journals, with promotional language in the conclusion. Many section of the paper are not properly formatted with overlapping texts and misaligned figures. For revision, I suggest completing citations, sharing code/datasets on GitHub, and adding ablation studies."*

Response:

1. **Excised Narrative Content:** Completely deleted 'The Story Behind ARES' and all journalistic/blog-style narratives. The manuscript opens with a formal academic Introduction focusing on technical problems, threat models, and architectural foundations.
2. **Objective Academic Tone:** Removed promotional superlatives ('revolutionary', 'unprecedented') from Abstract, Discussion, and Conclusion.
3. **Comprehensive Ablation Studies (Pages 24-37):** Added extensive multi-modal ablation matrices (Tables XXIII, XXV, XXIX, XXX, XXXII) evaluating system performance when individual modalities (vision, audio, behavioral bots, geopolitical context, temporal mutation tracking) are systematically removed ($p < 0.001$).

Comment 3.5: Real-World Use Cases, Human Study Ethics, and Grounded Benchmarking

> *"Use cases (e.g., disaster response, elections) highlight practical applicability."*

Response:

While practical applicability is a vital consideration, an objective re-assessment of our preliminary manuscript's Section XIII identified narrative claims of external societal impact:
- An assertion of a 47% reduction in panic traffic congestion during the 2022 Assam floods within 18 minutes.
- An assertion of an operational intelligence alert issued to military command during a border standoff.
- An assertion of automated intervention resulting in 13,000+ social media post removals.

- An assertion of an empirical public perception survey (n = 4,000) indicating a +62% trust gain.

In strict adherence to scientific reporting standards and research ethics:
1. **Complete Excision of Ungrounded Operational Claims:** We have completely excised all unsubstantiated operational claims regarding emergency deployment, military alerts, platform post removals, and public trust surveys from the text.
2. **Controlled Evaluation on ARES-Synth v1 [SIMULATED]:** The disaster, border conflict, and election scenarios were reframed into controlled, reproducible evaluations on the newly constructed **ARES-Synth v1** benchmark (Section XIII-A/B/D, Pages 13-14; Section XXVI-E, Page 28), explicitly labeled **[SIMULATED BENCHMARK EVALUATION]**:
- Track S1 (Border Conflict, n=620): Macro-F1 = 0.8982 [95% CI: 0.874, 0.921].
- Track S2 (Assam Floods, n=618): Macro-F1 = 0.9497 [95% CI: 0.931, 0.967].
- Track S3 (Electoral Disinformation, n=620): Macro-F1 = 0.8654 [95% CI: 0.838, 0.891].
3. **Empirical Retrospective Verification Traces:** In Section XIII-C and Section XIV, we provide empirical retrospective verification traces on real historical claims (CASE-01 to CASE-05) across frozen public corpora (CoAID, PubMed, X-FACT, AVeriTeC, FaceForensics++, ASVspoof 2019), documenting exact calibrated posterior probabilities P(False) and execution latencies (23.8-27.2 ms) without asserting live platform intervention.
4. **Institutional Review Board (IRB) Protocol Disclosure:** We explicitly state on Page 14 (Section XIII-E) that public perception surveys and user trust studies require formal Institutional Review Board (IRB) approval, designated for future clinical human-subjects trials.

5. Summary of Revisions by Manuscript Section and Page

The following table provides a complete correlation matrix mapping every reviewer comment to the corresponding page numbers, section headers, tables, and figures in the revised 43-page manuscript (ARES_Cognitive_Computation_Revised.pdf):

Manuscript Section	Page(s)	Key Revisions & Enhancements	Relevant Comments
Title & Front Matter	Page 1	Scoped academic title; author metadata formatted to publication standards; Abstract updated with bootstrap 95% CIs and 14.8-27.2 ms latency bounds.	Rev 1 (C1.4), Rev 2 (C2.2, C2.4), Rev 3 (C3.3, C3.4)
Section I: Introduction	Pages 1-3	Excised promotional narratives ('The Story Behind ARES'); formalized technical threat model and decision-theoretic scope.	Rev 3 (C3.4)
Section II-IV: Preprocessing	Pages 4-5	Vectorized Figure 3 (Throughput vs Concurrency) and Figure 6 (Timing Curves); O(n) algorithmic complexity derivations.	Rev 1 (C1.1, C1.3)
Section VII-IX: Bot Modeling	Pages 7-8	Vectorized Figures 10, 11, 12; Table VIII claim mutation cosine similarity logic.	Rev 1 (C1.3)
Section X-XI: Forensic Heads	Pages 9-11	Vectorized Figure 14 (GDELT ROC), Figure 15 (Ablation Matrix), Figure 16 (Calibration Curves); Table XI verified audio/deepfake metrics.	Rev 1 (C1.3), Rev 2 (C2.3)
Section XII: Threat Heatmap	Page 12	Vectorized Figure 19 (Regional Threat Heatmap).	Rev 1 (C1.3)
Section XIII: Use Cases & Impact	Pages 13-14	**Complete excision of operational deployment claims**; replaced with ARES-Synth v1 [SIMULATED] benchmarks and retrospective traces; clean 2D Figure 23.	Rev 1 (C1.3), Rev 3 (C3.5)
Section XIV: Ethical Safeguards	Pages 15-16	Table XVIII-B Ethical Safeguards Matrix; cryptographic SHA-256 hash-chained verification trail (O(1) append latency, 0.12 ms).	Rev 3 (Ethical Focus)
Section XV: Hardware & Latency	Page 17	**Table XX (RTX 4060 GPU Latency: 731.6 MB VRAM, 3,560 req/s)**; Table XXI Architectural Comparison; excision of ms/claim typo.	Rev 1 (C1.1, C1.2), Rev 3 (C3.1, C3.2)

Section XVI: Decision Formulation	Page 17	Formal loss function with risk-sensitive rejection threshold gamma(alpha) and cost ratio alpha = c_r / (c_a + c_r).	Rev 1 (C1.1), Rev 3 (C3.2)
Section XVII: Manifold Drift	Pages 18-19	Proposition 1 Bounded Mutation Detection; mathematical note on isotropic Gaussian noise vs anisotropic Riemannian correction.	Rev 1 (C1.1)
Section XIX-XXV: Complexity	Pages 20-23	Vectorized Figure 28 (Latency vs Tokens) and Figure 29 (Log-Log Scaling); Table XXII measured latencies.	Rev 1 (C1.1, C1.3)
Section XXVI: External Benchmarks	Pages 24-28	Evaluation on 13 external public benchmarks; Section XXVI-E ARES-Synth v1 Grounded Synthetic Benchmarks [SIMULATED].	Rev 1 (C1.4), Rev 3 (C3.3, C3.5)
Section XXVII-XXX: Ablations	Pages 24-26	Systematic modality ablation tables (Tables XXIII, XXV, XXIX, XXX).	Rev 3 (C3.4)
Section XXXI: Track Benchmarks	Pages 33-37	Table XXXI-B Calibrated Track Metrics (Table 7); Table XXXI-C Forensic Portfolio Analysis (Table 8) with explicit simulation disclosures.	Rev 2 (C2.3), Rev 3 (C3.2, C3.3)
Section XXXII: Walkthrough	Pages 34-35	**Page 34 raw listing replaced with Figure 38**; **Page 35 raw Mermaid code replaced with Figure 37**.	Rev 1 (C1.2)
Section XXXIII: Multi-Baseline	Page 40	**Table XXXVII empirical multi-baseline comparison**; **Algorithm 3 (Decision Abstention)** replacing duplicate algorithm.	Rev 1 (C1.2), Rev 3 (C3.2)
Appendices & Hyperparameters	Pages 41-43	Exhaustive training hyperparameters (Table XXXVIII); multi-task loss formulation with explicit line breaking; zero overflow.	Rev 1 (C1.1), Rev 3 (C3.2)
Bibliography	Pages 38-40	65 verified references with DOIs; zero [?] placeholders.	Rev 2 (C2.1), Rev 3 (C3.1)

We thank the Editor and the Reviewers once again for their invaluable guidance in elevating this manuscript. With all technical, empirical, and editorial concerns fully resolved, we believe ARES 2.0 represents a significant, rigorous, and reproducible contribution.